



COURSE DESCRIPTION FORM

INSTITUTION National University of Computer and Emerging Sciences (NUCES-FAST)
BS(SE)

PROGRAM (S) TO BE EVALUATED

A. Course Description

(Fill out the following table for each course in your computer science curriculum. A filled out form should not be more than 2-3 pages.)

Course Code	CS-2005
Course Title	Database Systems
Credit Hours	3+1
Prerequisites by Course(s) and Topics	CS-2001 (Data Structures)
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	Mid-I: 15 Mid-II: 15 Assignments/Quizzes: 10 Project: 10 Final: 50
Course Coordinator	Dr. Anam Qurashi
URL (if any)	
Current Catalog Description	Basic database concepts, Conceptual modelling, Relational data model, Relational theory and languages, Database design, SQL, Introduction to query processing and optimization, Introduction to concurrency and recovery with advance topics. This course provides students with the essential concepts, principles, and techniques of modern database systems from a user perspective. This means that the lecture focuses on the functionalities that are offered by database systems and not on the methods to implement them. Specifically, the course teaches students the ability to develop a solution for a real-world data management problem that requires the application of the theories and practices developed in class. From a theoretical point of view, this course covers the



	essential principles for the design, analysis, and use of computerized database systems. The design and techniques of conceptual modeling, database modeling, database system architecture, and user/program interfaces are presented in a unified way.																			
Textbook (or Laboratory Manual for Laboratory Courses)	amkant B. Navathe, <i>Database Systems, Models, Languages, Design and Application Programming</i> , 7 th Edition, 2016.																			
Reference Material	1) Thomas Connolly, Carolyn Begg, <i>Database Systems: A practical approach to design, implementation and Management</i> , 6 th Edition, 2015. 2) C.J. Date, <i>An Introduction to Database Systems</i> , 8 th Edition, 2004																			
Course Goals	<table border="1"> <tr> <th align="left" colspan="3">A. Course Learning Outcomes (CLOs)</th> </tr> <tr> <td colspan="3"> 1. Explain fundamental database concepts. 2. Analyze an information storage problem and derive an information model expressed in the form of an entity relation diagram and other optional analysis forms, such as a data dictionary. 3. Demonstrate an understanding of normalization theory to normalize the database and formulate, using SQL & relational algebra, solutions to a broad range of query & data problems in a team work. </td> </tr> <tr> <th align="left" colspan="3">B. Program Learning Outcomes</th> </tr> <tr> <td colspan="3">For each attribute below, indicate whether this attribute is covered in this course or not. Leave the cell blank if the enablement is little or non-existent.</td> </tr> <tr> <td>1. Academic Education:</td><td>To prepare graduates as computing professionals</td><td></td> </tr> <tr> <td>2. Knowledge for Solving Computing Problems:</td><td>Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.</td><td></td> </tr> </table>		A. Course Learning Outcomes (CLOs)			1. Explain fundamental database concepts. 2. Analyze an information storage problem and derive an information model expressed in the form of an entity relation diagram and other optional analysis forms, such as a data dictionary. 3. Demonstrate an understanding of normalization theory to normalize the database and formulate, using SQL & relational algebra, solutions to a broad range of query & data problems in a team work.			B. Program Learning Outcomes			For each attribute below, indicate whether this attribute is covered in this course or not. Leave the cell blank if the enablement is little or non-existent.			1. Academic Education:	To prepare graduates as computing professionals		2. Knowledge for Solving Computing Problems:	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.	
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	3. Problem Analysis:	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.	✓
	4. Design/ Development of Solutions:	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.	✓
	5. Modern Tool Usage:	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.	✓
	6. Individual and Team Work:	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings.	
	7. Communication:	Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.	
	8. Computing Professionalism and Society:	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.	



	9. Ethics:	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.																																																									
	10. Life-long Learning:	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.																																																									
	C. Relation between CLOs and PLOs (CLO: Course Learning Outcome, PLOs: Program Learning Outcomes) <table border="1"> <thead> <tr> <th colspan="2" rowspan="2"></th><th colspan="10">PLOs</th></tr> <tr> <th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th><th>9</th><th>10</th></tr> </thead> <tbody> <tr> <td rowspan="3">CLOs</td><td>1</td><td></td><td></td><td>✓</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>2</td><td></td><td></td><td></td><td>✓</td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>3</td><td></td><td></td><td></td><td></td><td>✓</td><td></td><td></td><td></td><td></td><td></td></tr> </tbody> </table>					PLOs										1	2	3	4	5	6	7	8	9	10	CLOs	1			✓								2				✓							3					✓					
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Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and one-hour lectures)	1. Topics to be covered: <table border="1"> <thead> <tr> <th>List of Topics</th><th>No. of Weeks</th><th>Contact Hours</th><th>CLO</th></tr> </thead> <tbody> <tr> <td>Chapter 1 Introduction, Characteristics of Database Approach, Files Vs. Databases, Characteristics of Database approach, Advantages of using DBMS, When not to use DBMS,</td><td>2</td><td>6</td><td>1</td></tr> <tr> <td>Chapter 2 Data Model, Schema and Instance, three schema architecture and data independence, classification of</td><td></td><td></td><td></td></tr> </tbody> </table>				List of Topics	No. of Weeks	Contact Hours	CLO	Chapter 1 Introduction, Characteristics of Database Approach, Files Vs. Databases, Characteristics of Database approach, Advantages of using DBMS, When not to use DBMS,	2	6	1	Chapter 2 Data Model, Schema and Instance, three schema architecture and data independence, classification of																																														
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	DBMS, database languages & Interfaces, Database systems environment.			
	Chapter 4 Relational Model Concepts, Relational Model Constraints			
	Chapter 5 Relational Database Schemas, Update Operations, Transactions, and Dealing with Constraint Violations			
	Chapter 6 SQL Data Definition and Data Types, Specifying Constraints in SQL, Basic Retrieval Queries in SQL, INSERT, DELETE, and UPDATE Statements in SQL, Additional Features of SQL	2	6	1,2
	Chapter 7 More Complex SQL Retrieval Queries, Views (Virtual Tables) in SQL, Schema Change Statements in SQL	1	3	1,2
	===== MID 1 =====			
	Chapter 3 Using High-Level Conceptual Data Models for Database Design, A Sample Database Application. Entity Types, Entity Sets, Attributes, and Keys, Relationship Types, Relationship Sets, Roles, and Structural Constraints, Weak Entity Types, Refining the ER Design for the COMPANY Database, ER Diagrams, Naming Conventions, and Design Issues, Relationship Types of Degree Higher than Two	1.5	4.5	2
	Chapter 8 Unary Relational Operations: SELECT and PROJECT Relational Algebra Operations from Set Theory Binary Relational Operations: JOIN and DIVISION	1	3	2



	Examples of Queries in Relational Algebra				
	Chapter 14				
	delines for				
	Relation Schemas				
	Functional Dependencies/Normal Forms Based on Primary Keys	2.5	7.5	3	
	General Definitions of Second and Third Normal Forms, Boyce-Codd Normal Form				
Multivalued Dependency and Fourth Normal Form					
Join Dependencies and Fifth Normal Form					
===== MID 2 =====					
Chapter 20					
Introduction to Transaction Processing					
Transaction and System Concepts					
Desirable Properties of Transactions					
Characterizing Schedules Based on Recoverability					
Characterizing Schedules Based on Serializability					
Transaction Support in SQL,					
	2	6	1,2		
Chapter 21					
Two-Phase Locking Techniques for Concurrency Control					
Concurrency Control Based on Timestamp Ordering					
Multiversion Concurrency Control Techniques					
Validation (Optimistic) Concurrency Control Techniques					
Granularity of Data Items and Multiple Granularity Locking					
Chapter 22					
Recovery Concepts					
NO-UNDO/REDO Recovery Based on Deferred Update	1.5	4.5	1,2		
Recovery Techniques Based on Immediate Update					
Chapter 24					
Introduction to NOSQL Systems					



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	Document-Based NOSQL Systems and MongoDB NOSQL Key-Value Stores NOSQL Systems				
	Review	0.5	1.5	1,2,3	
	Project Presentations	1	3	1	
	Total	15	45		
Laboratory Projects/Experiments Done in the Course					
Programming Assignments Done in the Course					
Class Time Spent on (in credit hours)	Theory	Problem Analysis	Solution Design	Social and Ethical Issues	
	30	10	5	0	
Oral and Written Communications	Every student is required to submit at least __1__ written report of typically __2__ pages and to make __1__ oral presentations of typically __10__ minute's duration. Include only material that is graded for grammar, spelling, style, and so forth, as well as for technical content, completeness, and accuracy.				

Instructor Name : Alina Arshad

Date : 19-08-2024